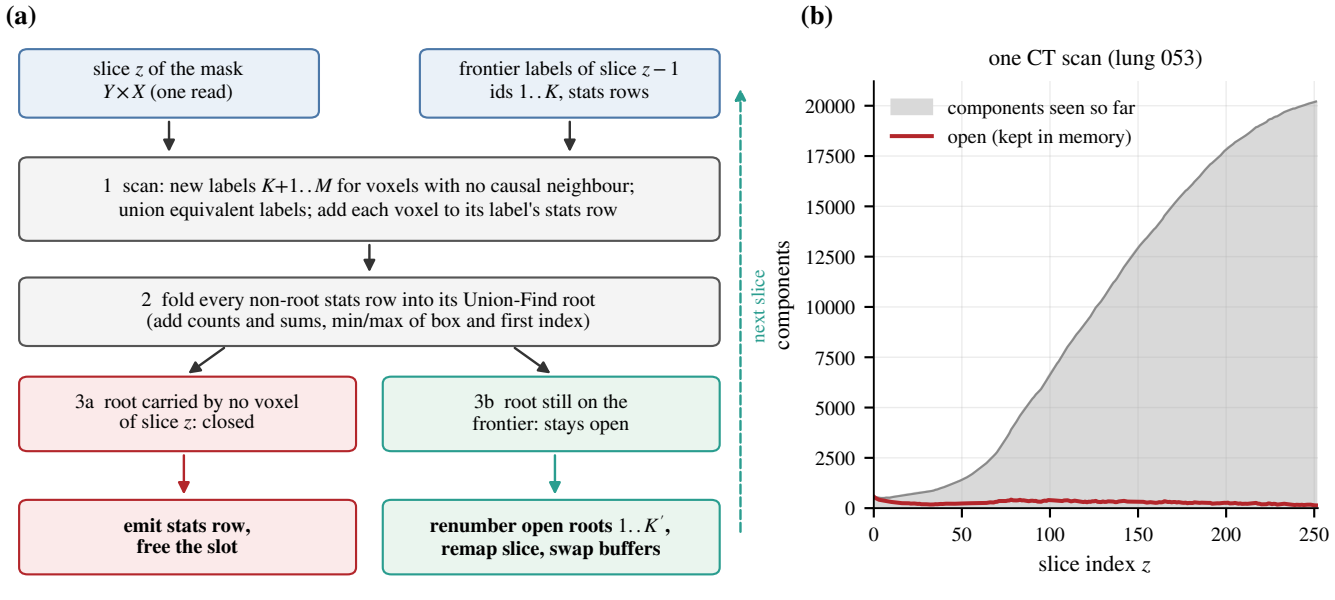
(a)
slice z of the mask
$Y \times X$ (one read)
frontier labels of slice $z-1$
ids $1..K$, stats rows
1 scan: new labels $K{+}1..M$ for voxels with no causal neighbour;
union equivalent labels; add each voxel to its label's stats row
2 fold every non-root stats row into its Union-Find root
(add counts and sums, min/max of box and first index)
3a root carried by no voxel
of slice z: closed
3b root still on the
frontier: stays open
emit stats row,
free the slot
renumber open roots $1..K'$,
remap slice, swap buffers
next slice
(b)
one CT scan (lung 053)
components seen so far
open (kept in memory)
components
20000
17500
15000
12500
10000
7500
5000
2500
0
0
50
100
150
200
250
slice index z